Prerequisites

Def (Join and Meet)**.**
- ***Join*** *$a = \vee S$ if a upper bound and if b upper bound of S, then $a \leq b$. Colimit*
- ***Meet*** *$a = \wedge S$ if a lower bound and if b lower bound of S, then $b \leq a$. Limit*

Def (Lattice)**.** *Poset L in which every finite subset has a join and meet.*
- ***Distributive*** *if $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$. Morphisms preserve finite joins and meets. Category* Dlat.
- ***Complete*** *if arbitrary subsets have a join and meet.*
- ***Frame*** *Complete and $f \wedge (\vee S) = \vee (f \wedge s : s \in S)$. Morphisms preserve arbitrary joins and finite meets. Category* Frm
- ***Finite*** *$c \in F$ if $c \leq \vee S$ implies $c \leq \vee K$ for some finite $K \subseteq S$. F^ω set of finite elements.*
- ***Coherent Frame*** *Every element is a join of finite elements and F^ω is distributive lattice. Morphism take $F^\omega \to G^\omega$. Category* CFrm.

Prop. *Adjunction $f : X \to Y$, $\mathcal{O}(f) : \mathcal{O}(Y) \to \mathcal{O}(X)$ such that $U \mapsto f^{-1}(U)$ (explain better the pt part)*

$$\mathrm{Frm}^{op} \underset{\mathcal{O}}{\overset{pt}{\rightleftarrows}} \mathrm{Top}$$

Prop. *Equivalence of categories*

$$\mathrm{Dlat} \underset{(-)^\omega}{\overset{\mathrm{Idl}}{\rightleftarrows}} \mathrm{CFrm} \quad (\sim)$$

Def (Sober space)**.** *T_0 and every irreducible closed subset is the closure of a unique point.*

Def (Spectral space)**.** *qc sober space X qc opens are closed under finite intersections and form a basis of the topology. Morphisms are continuous maps such that the preimage of a qc open is qc open. Category* TopSpec.

Theorem (Stone duality)**.** *Equivalence of categories*

$$\mathrm{CFrm}^{op} \underset{\mathcal{O}}{\overset{pt}{\rightleftarrows}} \mathrm{TopSpec} \quad (\sim)$$

Def (Hochster duality)**.** *∂ changes the orientation of the order.*
- *X^∂ same set, but basis of closed sets is the basis of qc opens of X. Open sets of X^∂ are Thomason, $V = \cup V_\lambda$ with V_λ closed and qc complement.*
- $\mathrm{Th}(X) = \mathcal{O}(X^\partial)$, $\quad \mathrm{Th}(X^\partial) = \mathcal{O}(X)$

TT-geometry

Def. *$\mathcal{K}$ essentially small tt-cat.*
- *$\mathcal{I} \subseteq \mathcal{K}$ full is* ***thick*** *if it is closed under suspension, cones and direct summands.*
- *$\mathcal{I} \subseteq \mathcal{K}$ thick is* ***ideal*** *if $i \otimes k \in \mathcal{I}$ for all $i \in \mathcal{I}$ and $k \in \mathcal{K}$.*
- *$\mathcal{I} \subseteq \mathcal{K}$ ideal is* ***radical*** *if $k^{\otimes n} \in \mathcal{I}$ for some $n > 0$ implies $k \in \mathcal{I}$.*
- $\mathrm{Zar}(\mathcal{K})$ *set of radical ideals of $\mathcal{K}$.*
- *$\mathfrak{p} \subseteq \mathcal{K}$ prime ideal if proper and $k \otimes k' \in \mathfrak{p}$ implies $k \in \mathfrak{p}$ or $k' \in \mathfrak{p}$.*

Prop. $\sqrt{}$ *prime* $\Rightarrow$ $\sqrt{}$ *radical.*

Def. *Let $a \in \mathcal{K}$. Notation for generation of subcategories / ideals*

$$\mathrm{thick}\langle a\rangle, \quad \mathrm{thick}_\otimes\langle a\rangle, \quad \sqrt{a}$$

Theorem. $\mathrm{Zar}(K)$ *is a coherent frame.*
- $\mathcal{I} \wedge \mathcal{J} = \mathcal{I} \cap \mathcal{J} = \sqrt{\mathcal{I} \otimes \mathcal{J}}$
- $\mathcal{I} \vee \mathcal{J} = \sqrt{\mathcal{I} \cup \mathcal{J}} = \sqrt{\mathcal{I} \oplus \mathcal{J}}$

Def (Balmer Spectrum)**.** $\mathrm{Spc}(\mathcal{K}) \coloneqq \mathrm{pt}(\mathrm{Zar}(\mathcal{K}))^\partial = \mathrm{pt}(\mathrm{Zar}(\mathcal{K})^\partial)$

Prop. *There is a lattice isomorphism* $\mathrm{Th}(\mathrm{Spc}(\mathcal{K})) \cong \mathrm{Zar}(\mathcal{K})$ *st*

$$\{\textit{Thomason subsets of } \mathrm{Spc}(\mathcal{K})\} \leftrightarrow \{\textit{radical ideals of } \mathcal{K}\}$$

Prop. $\{$*points in the frame* $\mathrm{Zar}(\mathcal{K})\} \cong \{$*prime ideals of* $\mathcal{K}\}$

Def (Support)**.** *For $a \in \mathcal{K}$, define* $\mathrm{supp}(a) \coloneqq \{\mathfrak{p} \in \mathrm{Spc}(\mathcal{K}) : a \notin \mathfrak{p}\}$ *Complement $U(a)$*

Prop. *$U(a)$ is a basis of quasi-compact opens of* $\mathrm{Spc}(\mathcal{K})$ *and* $\mathrm{supp}(a)$ *is a basis of closed subset (Thomason subsets).*
- $\mathrm{supp}(0) = \emptyset$, $\mathrm{supp}(1) = \mathcal{K}$
- $\mathrm{supp}(a \oplus b) = \mathrm{supp}(a) \cup \mathrm{supp}(b)$, $\mathrm{supp}(a \otimes b) = \mathrm{supp}(a) \cap \mathrm{supp}(b)$
- $\mathrm{supp}(\Sigma a) = \mathrm{supp}(a)$
- $\mathrm{supp}(b) \subseteq \mathrm{supp}(a) \cup \mathrm{supp}(c)$ *for any triangle $a \to b \to c$ in $\mathcal{K}$*

Def (Support datum)**.** *Pair (X, σ) with X a topological space and $\sigma : \mathcal{K} \to$ closed subsets of X acomplishing the same properties as avobe.*

Theorem (Universal property)**.** $(\mathrm{Spc}(\mathcal{K}), \mathrm{supp})$ *is the final support datum. For any other (X, σ), there exist a unique continuous map $f : X \to \mathrm{Spc}(\mathcal{K})$ such that $\sigma(a) = f^{-1}(\mathrm{supp}(a))$ for all $a \in \mathcal{K}$.*